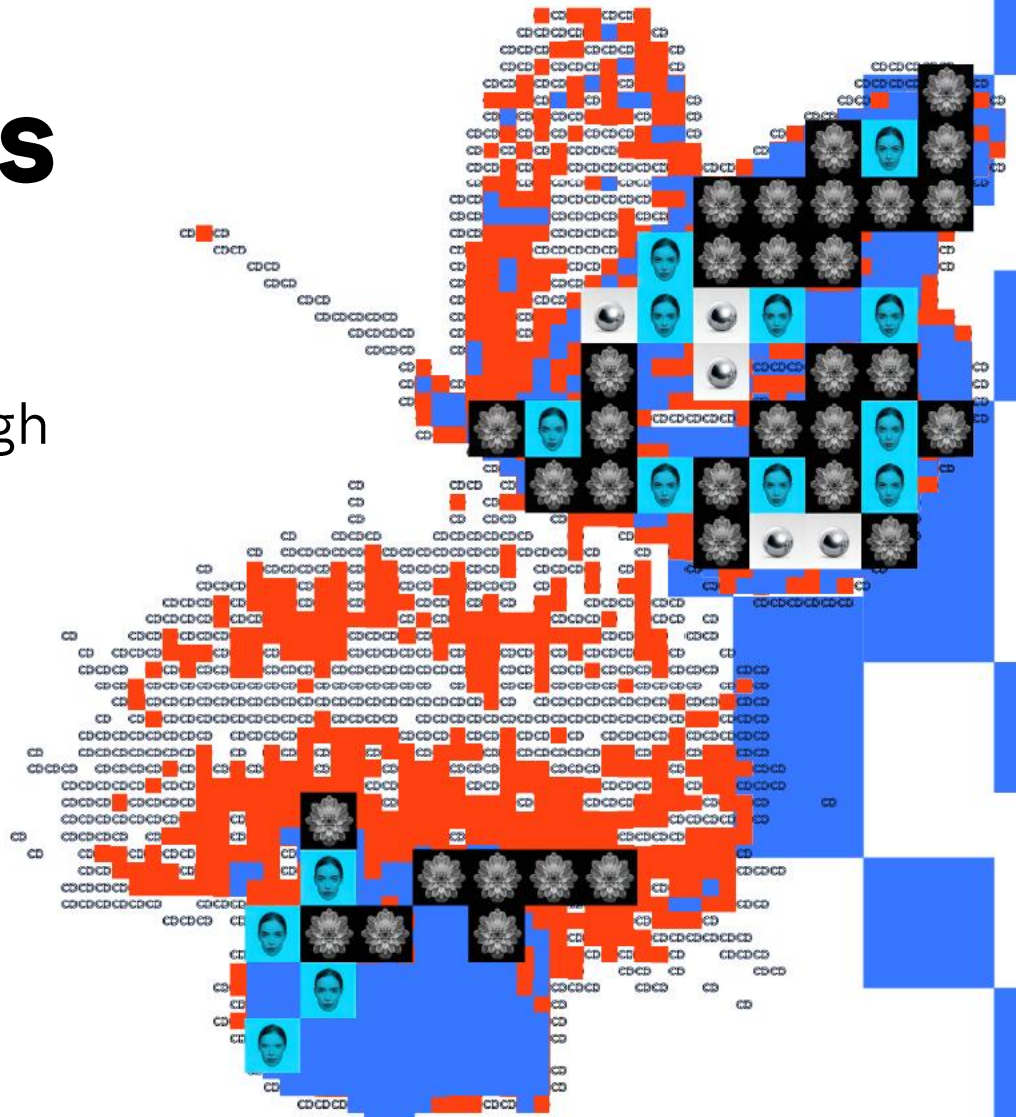
NOVA IMS INFORMATION MANAGEMENT SCHOOL

Big Data Analysis in Spark

Optimizing Airline Operations Through Scalable Data Insights



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Introduction

The Cost of Delay

In the world of commercial aviation, delays represent massive financial losses and frustration for millions of passengers. In 2023, flight delays in the US cost the industry a lot of money in wasted fuel, compensation, and lost productivity. Our challenge is to transform this cost into Operational Intelligence.

Our Approach

Traditional methods fail because they treat delays as **isolated events**. In reality, a delay at one airport propagates systemically throughout the network. Our goal: Shift the focus from "Where is the delay?" to **"How does the delay propagate?"** and **"Where is the network most vulnerable?"**

Our Dataset

We utilized a public, granular dataset from the US Department of Transportation (DOT), containing **over 17 million records** of delays and flights from 2023. This volume of data is only processable and analyzable using a Big Data architecture.

Operational Health Check (RDDs)

Volume, Hubs & Outliers

Total flights in dataset: 171666

Top 5 Airlines by Accumulated Delay

```
Southwest Airlines Co.: 2,460,563.00 Minutes
American Airlines Inc.: 1,538,261.00 Minutes
Delta Air Lines Inc.: 1,198,884.00 Minutes
SkyWest Airlines Inc.: 1,167,810.00 Minutes
United Air Lines Inc.: 1,030,741.00 Minutes
```

Outliers: Flights Exceeding Airport Average

carrier	airport	arr_delay	airport_avg_delay	deviation_from_avg
AA	DFW	438783.0	26513.337217771303	412269.66278222867
DL	ATL	429194.0	23149.15165562914	406044.84834437084
AA	DFW	416577.0	26513.337217771303	390063.66278222867
DL	ATL	395609.0	23149.15165562914	372459.84834437084
AA	DFW	376057.0	26513.337217771303	349543.66278222867

airport	avg_delay_min
ORD	28688.19
DFW	26513.34
ATL	23149.15
DEN	21660.03
SFO	19627.08

Top 5 Airports by Average Delay

Exploratory Analysis

Operational Reality Check (EDA)

- **Seasonality:** Identified July as the peak risk month for weather disruptions, contradicting winter assumptions.
- **Propagation:** Validated strong correlation between *Carrier Delays* and *Late Aircraft*, confirming the "Snowball Effect" hypothesis.

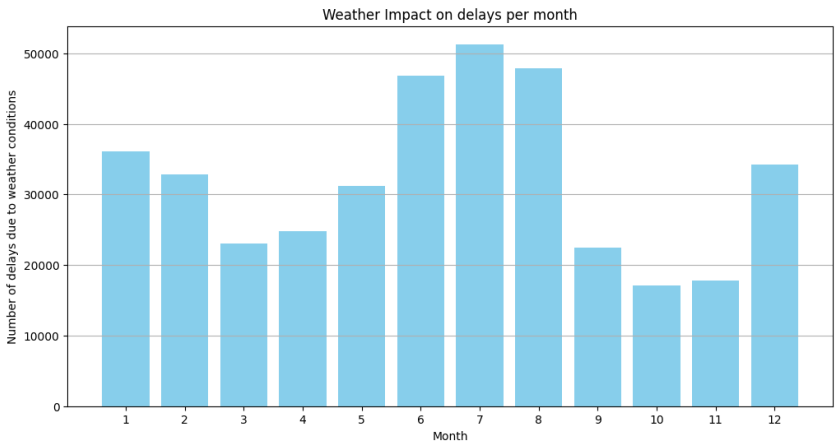


Figure 1 – Weather impact on delays per month

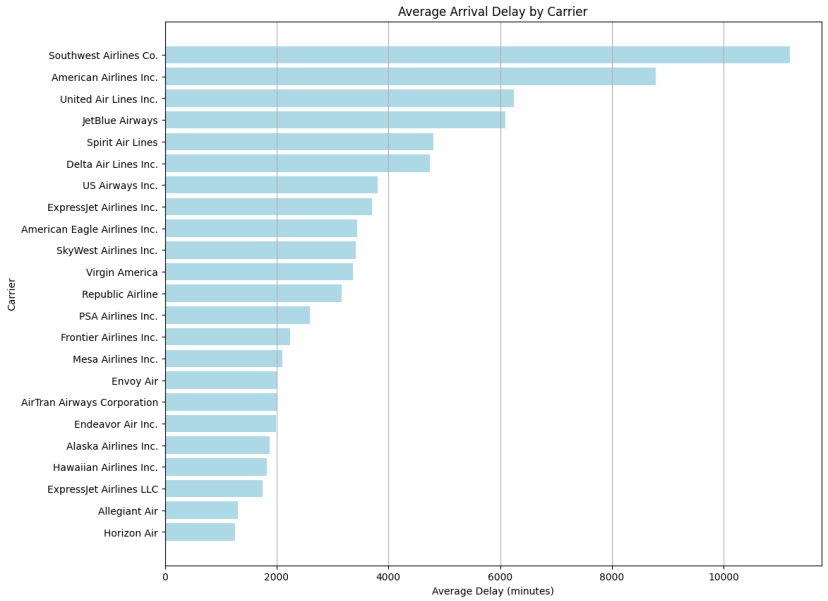


Figure 2 – Average Arrival Delay per carrier

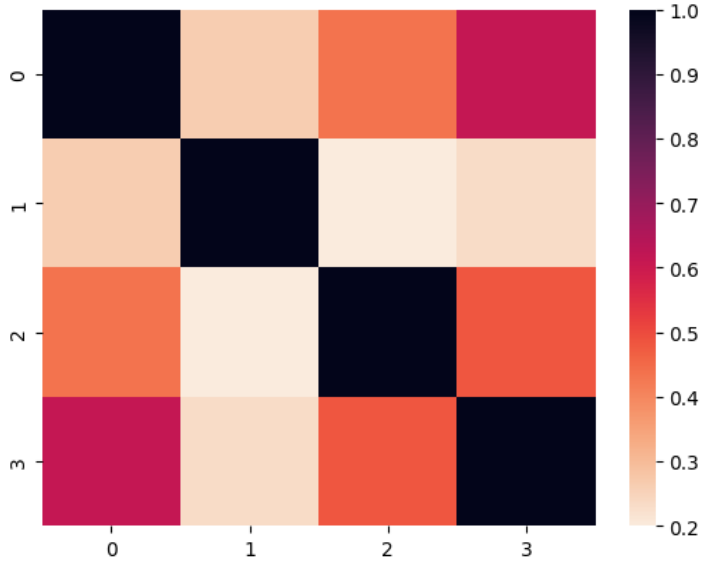


Figure 3 – Correlation Analysis

Strategic Modeling Decisions

1. Strategic Target Engineering: Defining "True Risk"

- Action: We engineered a custom binary variable, "***significative_delay***" , which flags instances where the delay ratio exceeds 15% of total operations.

- We deliberately chose *not* to predict minor, negligible delays (e.g., a single flight arriving 5 minutes late), as this is merely operational noise. By training the model to detect only Systemic Collapses (>15% failure rate), we reduce false alarms and ensure management focuses exclusively on critical days that require intervention.

2. Statistical Noise Reduction (Outlier Removal)

- Action: Applied the Interquartile Range (IQR) method to mathematically identify and remove extreme data points before training.

- filtered out Non-Recurring extreme events, such as rare hurricanes or global IT outages, that skew the data. This ensures the model learns the repeatable patterns of daily operations rather than memorizing anomalies. The result is a robust planning tool for standard day-to-day management, rather than a disaster-response log.

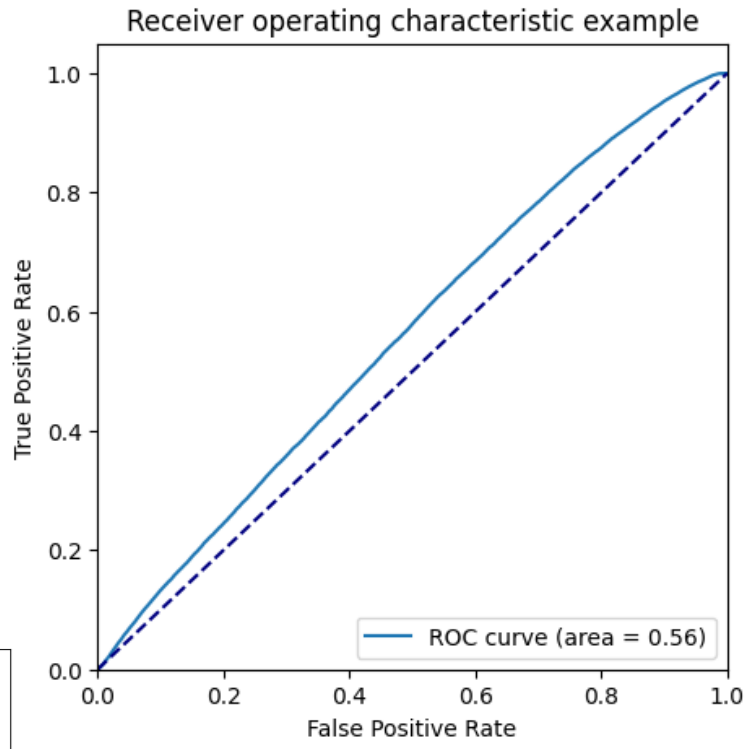
Feature Engineering and Predictive Model Selection

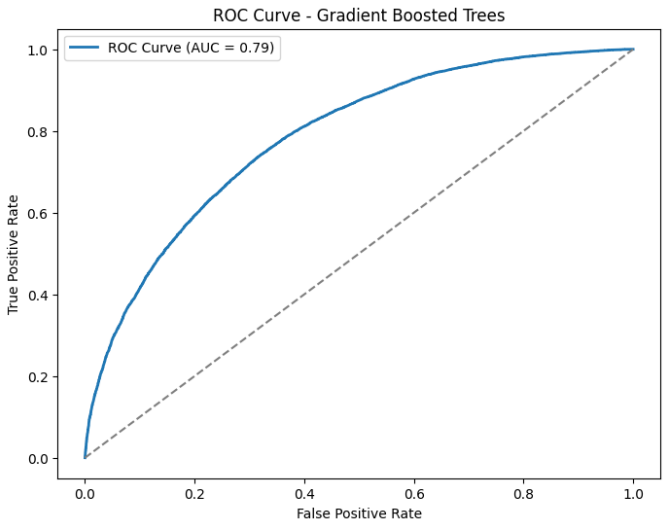
The Modeling Breakthrough (Model Selection)

The Challenge: Linear models (Logistic Regression) failed to classify "Significant Delays" (AUC 0.56), **unable to distinguish between actual risks (True Positives) and false alarms (False Positives).**

Solution: Leveraged **Gradient Boosted Trees (GBT)** using Spark MLlib on our engineered features.

The Result: Achieved a **79% AUC**, proving the system can reliably **filter out False Positives and accurately identify True Positives** (actual systemic breakdowns).





Automated Data Pipeline & Scalable Architecture

01

Medallion Architecture

Bronze: Raw data ingestion without information loss (Auditability)

Silver: Automatic error cleaning and null value handling (Data Quality).

Gold: Data ready for consumption by ML and Management Dashboards.

Integration of the Predictive Model into the Automated Pipeline

02

03

Strategic Finding: The Volume Paradox

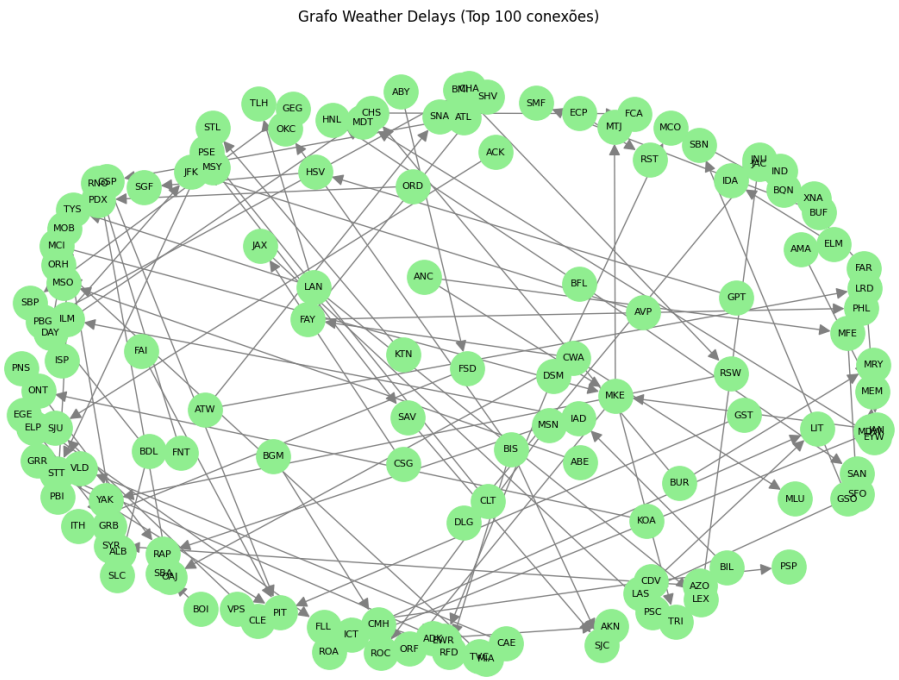
Pipeline analysis revealed that **Flight Volume** is NOT a primary driver of systemic delays (Coefficient ≈0).
Conclusion: Operational efficiency, not just capacity, is the key leverage point for reducing delays.

Identifying structural points of failure within the aviation network.

The Model: Mapping "Delay Contagion", Not Just Flights
Standard maps show where planes fly. We built a network to track where delays travel. If a carrier is late in Airport A and flies to Airport B, we created a "Contagion Link". We weighted these links by Total Delay Minutes. The network highlights the heavies waste of time, ignoring healthy, on-time traffic.

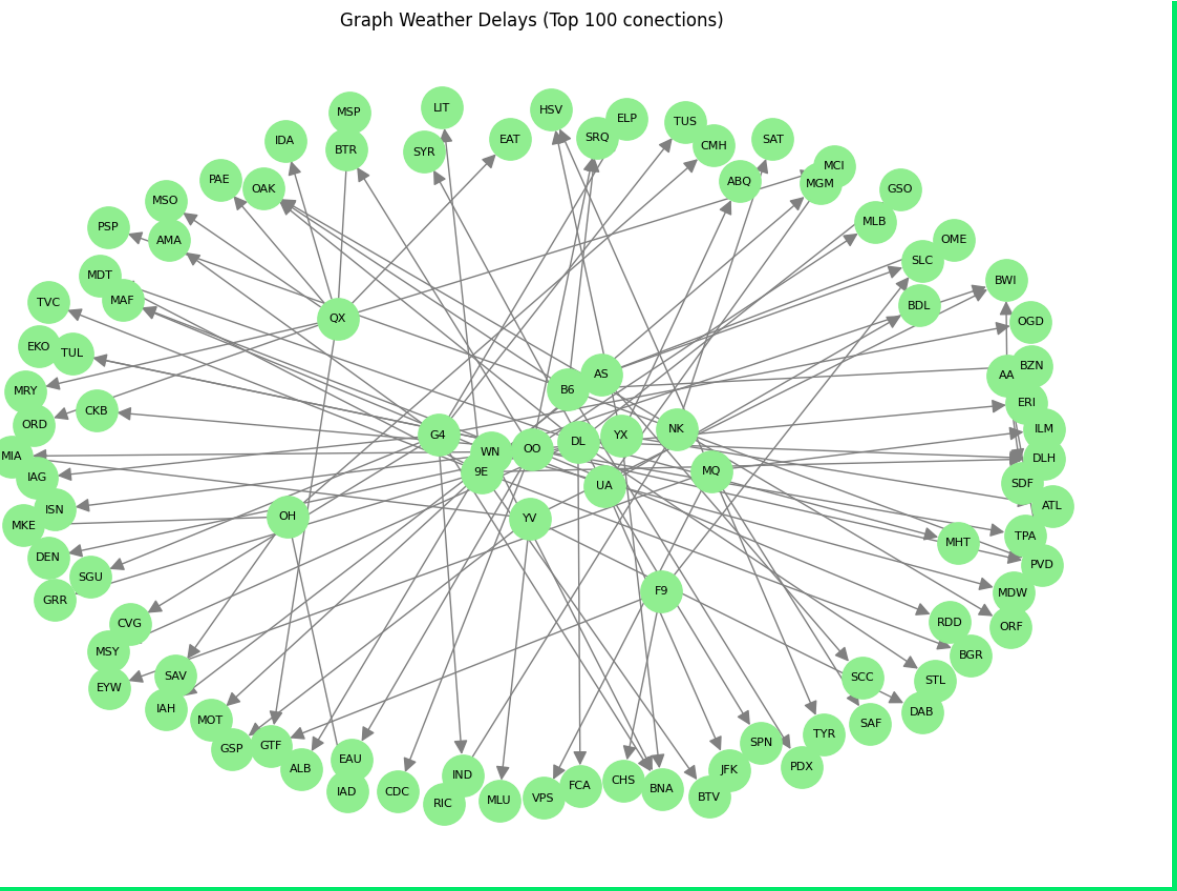
Diagnosis: Simple volume and true systemic risk.
At high-traffic airports like Binghamton, delays remain locally contained, Yuma International functions as the network's primary 'Super-Spreader.' Despite its smaller size, Yuma feeds directly into major arteries, meaning a delay there triggers a domino effect that destabilizes hubs across the entire country.

Actionable Alert: The 12-Million-Minute Corridor
We isolated specific routes that are bleeding efficiency.
The Smoking Gun: The corridor Atlanta (ATL) → Greenville (GSP) is a structural failure, accumulating over 12 Million minutes of carrier delay.
This is not a random fluctuation. It is a specific, repetitive operational breakdown on this route.



Graph 1 -Network Analysis - Carrier Delays

Propagation Analysis: Delay Tracking & Mitigation



Graph 2 - Weather Delays

The Model: Mapping Pressure Points

Weather disrupts Carrier Fleets, which then collapse specific airports. We modeled the pressure that Carriers exert on Airports, tracking the total weather delay transferred from specific Airlines to specific Hubs

No Single Culprit (Systemic Saturation)

•Contrast: Unlike Graph 1 (where Yuma was the problem), Weather risk is evenly distributed. PageRank scores are flat (~1.16). Even the top affected airport (Austin/AUS) is barely distinguishable from the rest.

You cannot "build" your way out of weather delays. Infrastructure upgrades at one airport won't fix this, because the risk is spread across the entire top tier.

Actionable Alert: The "Broken Links" Strategy

Since we can't fix the weather or a single hub, we must fix Specific Operational Pairs.

We identified massive failures in specific combinations, such as Delta (DL) to St. Louis (STL) and Delta to New Orleans.The failure is the **link**, not the **location**.

Thank you

Dataset

Welcome To Zscaler Directory Authentication. (2025). Kaggle.com. https://www.kaggle.com/datasets/sriharshaeedala/airline-delay/data